RATE ANALYSIS (2082/083)

Description of works:	Providing and laying of Reinforced / Pre-stressed Cement Concrete in Solid Slab Super Structure as per Drawing and Technical Specifications. RCC Grade M25 Height above 5 to 10m (including formworks)	Unit : 15 m3
Spec. cl. No: 2000		

Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
20.8 B (i)(q)	Skilled	md	3.00	1215.00	3645.00	cement	mt	5.990	13500.00	80865.00	mixer (0.28/ 0.20 m^3)	hr	6.00	826.26	4957.56
	Unskilled	md	30.00	855.00	25650.00	aggregate 20mm	m^3	8.100	1990.00	16119.00	Generator	hr	6.00	1173.44	7040.64
						aggregate 10mm	m^3	5.400	2044.00	11037.60					
						Coarse Sand	m^4	6.750	2329.00	15720.75					
						water	lit	3000.000	0.19	577.50					
						Admixture @ 0.4 % of Cement	KG	23.960	288.00	6900.48					
				Sub total of A =	29295.00	Formwork & Staging @25 % of A+B+C			Sub total of B =	131220.33				Sub total of C =	11998.20
				Sub total of A +B + C =	172513.53			43128.38	Contractor's overhead expenses 15% =		32346.29			Total Rate =	247988.20
														Rate Per 1m3	16532.55

Description of works:	Providing and laying of Reinforced / Pre-stressed Cement Concrete in Solid Slab Super Structure as per Drawing and Technical Specifications. RCC Grade M25 Height above 10m (including Formworks)	Unit : 15 m3
Spec. cl. No: 2000		

Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
20.8 B (i)(r)	Skilled	md	3.00	1215.00	3645.00	cement	mt	5.990	13500.00	80865.00	mixer (0.28/ 0.20 m^3)	hr	6.00	826.26	4957.56
	Unskilled	md	30.00	855.00	25650.00	aggregate 20mm	m^3	8.100	1990.00	16119.00	Generator	hr	6.00	1173.44	7040.64
						aggregate 10mm	m^3	5.400	2044.00	11037.60					
						Coarse Sand	m^4	6.750	2329.00	15720.75					
						water	lit	3000.000	0.19	577.50					
						Admixture @ 0.4 % of Cement	KG	23.960	288.00	6900.48					
				Sub total of A =	29295.00	Formwork & Staging @30 % of A+B+C			Sub total of B =	131220.33				Sub total of C =	11998.20
				Sub total of A +B + C =	172513.53			51754.06	Contractor's overhead expenses 15% =		33640.14			Total Rate =	257907.73
														Rate Per 1m3	17193.85

Description of works:	Providing and laying, fitting and placing HYSD bar reinforcement in super-structure complete as per Drawing and Technical Specifications	Unit : 1 ton.
Spec. cl. No: 2014		

Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
20.9	Skilled	md	4.00	1215.00	4860.00	Reinforcement	t	1.100	91000.00	100100.00					
	Unskilled	md	12.00	855.00	10260.00	Binding wire	kg	8.000	111.00	888.00					
				Sub total of A =	15120.00				Sub total of B =	100988.00				Sub total of C =	0.00
				Sub total of A +B + C =	116108.00				Contractor's overhead expenses 15% =		17416.20			Unit Rate =	133524.20
														Rate Per KG	133.52

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Description of works:	Provide mechanically woven double twisted crates/mattress including rolling cutting and with lacing wire and binding wire as per specification	Unit : 1 Sqm
Spec. cl. No: 2402		

Norms No. 24.2	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
						Machine made gabion	sqm	1.00	310.20	310.20					
				Sub total of A =	0.00				Sub total of B =	310.20				Sub total of C =	0.00
				Sub total of A +B + C =	310.20				Contractor's overhead expenses 15% =		46.53			Unit Rate =	356.73

Description of works:	Assembling mechanically woven gabion boxes/ mattresses placing in position including streching forming compartments, tying the sides and diapharagm with binding wire in each mesh tying the bracing wire and tie wires and tying down the lid complete as per specification	Unit : 160 sqm
Spec. cl. No: 2401		

Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
24.3	Skilled	md	2.00	1215.00	2430.00										
	Unskilled	md	10.00	855.00	8550.00										
				Sub total of A =	10980.00				Sub total of B =	0.00				Sub total of C =	0.00
				Sub total of A +B + C =	10980.00				Contractor's overhead expenses 15% =		1647.00			Unit Rate =	12627.00
														Rate per m2	78.92

Description of works:	Providing and filling stone/ boulder in gabion boxes/ mattresses etc including dressing, bedding, bonding all complete as per drawing and technical specification	Unit : 10 m3
Spec. cl. No: 2401		

Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
24.4	Skilled	md	2.00	1215.00	2430.00	Boulder/Stone	m3	11.00	2218.00	24398.00					
	Unskilled	md	8.00	855.00	6840.00										
				Sub total of A =	9270.00				Sub total of B =	24398.00				Sub total of C =	0.00
				Sub total of A +B + C =	33668.00				Contractor's overhead expenses 15% =		5050.20			Unit Rate =	38718.20
														Rate per m3	3871.82

24.3	Supply of machine made hexagonal mesh type 100mm x 120 mm Gabion boxes / mattresses with diapharagms, with binding wire 12 swg, mesh wire 3mm and selvedged wire 3.9 mm and								Unit : 1 m3
	Box size	Unit	Sqm	Filling Cum	Supply of box	Assembling	Filling	Total	Per m3
	1.5 x 1 x 1		8	1.50	2853.84	631.35	5807.73	9292.92	6195.28
	2 x 1 x 1		11	2.00	3924.03	868.11	7743.64	12535.78	6267.89
	3 x 1 x 1		16	3.00	5707.68	1262.70	11615.46	18585.84	6195.28
						Average per m3			6219.48	6219.48

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Description of works:	Providing, laying and fixing of Geo-textile (filter fabrics), lead 30m	Unit :300 m2
Spec. cl. No: 2404		

Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
24.5	Unskilled	md	1.00	855.00	855.00	Geotextile	m2	360.00	117.14	42170.40					
	Skilled	md	2.00	1215.00	2430.00										
				Sub total of A =	3285.00				Sub total of B =	42170.40				Sub total of C =	0.00
				Sub total of A +B + C =	45455.40				Contractor's overhead expenses 15% =	6818.31				Unit Rate :	52273.71
														Rate per m2	174.25

Description of works:	Providing and laying Plum Concrete (Boulder Mixed Concrete) as per Drawing and Specifications 60% M15 concrete and 40% boulders/stone using Mechanical Means.(including formworks)	Unit : 10 m3
Spec. cl. No: 2421		

Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
24.21	Skilled	md	3.00	1215.00	3645.00	cement	mt	1.700	13500.00	22950.00	mixer (0.28/ 0.20 m^3)	hr	6.00	826.26	4957.56
	Unskilled	md	30.00	855.00	25650.00	aggregate(20- 40)mm	m^3	3.450	1990.00	6865.50	Concrete Vibrator	hr	6.00	214	1284.00
						aggregate (10-20)mm	m^3	1.560	1990.00	3104.40					
						aggregate(5-10)mm	m^3	0.720	2044.00	1471.68					
						Sand	m^3	3.000	2329.00	6987.00					
						Boulder Stone	m^3	4.400	2218.00	9759.20					
				Sub total of A =	29295.00				Sub total of B =	51137.78				Sub total of C =	6241.56
				Sub total of A +B + C+d =	86674.34				Contractor's overhead expenses 15% =	13001.15				Total Rate =	99675.49
														Rate per m3	9967.55

Description of works:	Filling Pot-holes and Patch Repairs with open-Graded Premix surfacing, 20 mm. Providing required material and repair the the pot holes including removal of failed material, trimming and finishing the surface applying tack coat on the sides and base of excavation , backfilling with hot bituminous Material andcompaction as per Technical Specification and instruction of the Engineer.	Unit : 600 Sqm.
Spec. cl. No: 2903		

Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
29.11 A	Skilled	md	1.00	1215.00	1215.00	Bitumen	kg	1080.00	110.00	118800.00	Hot Mix Plant	hr	1.00	5690.02	5690.02
	Unskilled	md	6.00	855.00	5130.00	Crushed stone aggregates nominal size 13.2 mm	Cum	12.96	1990.00	25790.40	Air Compressor	hr	6.00	1727.47	10364.82
						Crushed stone aggregates nominal size 5mm	Cum	6.48	2044.00	13245.12	Tripper/Tractor	hr	6.00	1283.38	7700.28
						Bitumen or emulsion (for Prime and Tack Coat)	Kg	360.00	92.84	33422.40	Smooth wheeled roller	hr	6.00	1875.20	11251.20
				Sub total of A =	6345.00				Sub total of B =	191257.92				Sub total of C =	35006.32
				Sub total of A +B + C =	232609.24				Contractor's overhead expenses 15% =	34891.39				Total	267500.63
														Unit Rate	445.83

Description of works	Planting rooted grass slips on the slopes <45o including preparation of slips on site. a max of 5 cm depth with metal rod or Operation includes digging planting hole to hard-wood peg, depending on the nature of the soil. The planting drills should be space	Unit = sqm (For 1 sqm)
Spec. cl. No: 2807		

Norms no 28.08 E	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
	Unskilled	day	0.20	855.00	171.00	Grass slips/ no of drills Cuttings	nos	100	3.30	330.00	Add 3 % of Labour cost for Ms rod or hard wood peg and other T&P			5.13	5.13
						Hessian Jute	sqm	0.27	60.00	16.20					
	Sub total A				171.00	Sub total of B =				346.20	Sub total of C =				5.13

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	Sub total of A +B + C =				522.33	Contractor's overhead expenses 15% =				78.35	Total				600.68		
															Unit Rate		600.68
Description of works Spec. cl. No: 2807	Planting rooted grass slips on the slopes 45o - 60o including preparation of slips on site. a max of 5 cm depth with metal rod or Operation includes digging planting hole to hard-wood peg, depending on the nature of the soil. The planting drills should be space																Unit = sqm (For 1 sqm)
Norms no 28.08 F	Labour (A)					Material (B)					Equipment (C)						
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount		
	Unskilled	day	0.30	855.00	256.50	Grass slips/ no of drills Cuttings	nos	100	3.30	330.00	Add 3 % of Labour cost for Ms rod or hard wood peg and other T&P			7.70	7.70		
						Hessian Jute	sqm	0.27	60.00	16.20							
	Sub total A				256.50	Sub total of B =				346.20	Sub total of C =				7.70		
	Sub total of A +B + C =				610.40	Contractor's overhead expenses 15% =				91.56	Total				701.95		
															Unit Rate		701.95
Description of works Spec. cl. No: 2807	Planting rooted grass slips on the slopes > 60o including preparation of slips on site. a max of 5 cm depth with metal rod or Operation includes digging planting hole to hard-wood peg, depending on the nature of the soil. The planting drills should be space																Unit = sqm (For 1 sqm)
Norms no 28.08 G	Labour (A)					Material (B)					Equipment (C)						
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount		
	Unskilled	day	0.40	855.00	342.00	Grass slips/ no of drills Cuttings	nos	100	3.30	330.00	Add 3 % of Labour cost for Ms rod or hard wood peg and other T&P			10.26	10.26		
						Hessian Jute	sqm	0.27	60.00	16.20							
	Sub total A				342.00	Sub total of B =				346.20	Sub total of C =				10.26		
	Sub total of A +B + C =				698.46	Contractor's overhead expenses 15% =				104.77	Total				803.23		
															Unit Rate		803.23
Description of works Spec. cl. No: 2803	Collection of hardwood for cuttings for planting Material (e.g. assuro, namdi phul, simali)from the sources within of the road. Material to be minimum. 30cm in length and minimum 2 cm in diameter																Unit = nos. (For 100 nos.)
Norms no 28.02 C	Labour (A)					Material (B)					Equipment (C)						
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount		
	Unskilled	day	0.85	855.00	726.75	Hessian Jute	sqm	5	60.00	300.00	Add 3 % of Labour cost for Ms rod or hard wood peg and other T&P			21.80	21.80		
										-							
	Sub total A				726.75	Sub total of B =				300.00	Sub total of C =				21.80		
	Sub total of A +B + C =				1,048.55	Contractor's overhead expenses 15% =				157.28	Total				1,205.84		
															Unit Rate		12.06
Description of works Spec. cl. No: 2805	Mulch production by collection and cutting of weeds and other vegetation such as tite pati, banmara etc., within 1 km of the road, and stacking along roadside.																Unit = cum. (For 1 cum.)
Norms no 28.6A	Labour (A)					Material (B)					Equipment (C)						
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount		
	Unskilled	day	0.12	855.00	102.60					-	Add 3 % of Labour cost for Ms rod			3.08	3.08		
										-							
	Sub total A				102.60	Sub total of B =				-	Sub total of C =				3.08		
	Sub total of A +B + C =				105.68	Contractor's overhead expenses 15% =				15.85	Total				121.53		
															Unit Rate		121.53
Description of works Spec. cl. No: 2805	Compost production by collection and cutting of weeds and other vegetation such as tite pati, banmara etc., within 1 km of the road, including fine cutting and filling compost pit.																Unit = cum. (For 1 cum.)
Norms no 28.6B	Labour (A)					Material (B)					Equipment (C)						
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount		
	Unskilled	day	0.12	855.00	102.60					-	Add 3 % of Labour cost for Ms rod			3.08	3.08		
										-							
	Sub total A				102.60	Sub total of B =				-	Sub total of C =				3.08		
	Sub total of A +B + C =				105.68	Contractor's overhead expenses 15% =				15.85	Total				121.53		
															Unit Rate		121.53

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Description of works	Planting containerized tree and shrub seedlings, including pitting, transplanting, composting and mulching, on slopes < 30o . Pit size 30 cm diameter x depth mix Compost with soil and back fill into pit to 1/4 of the pit volume														Unit = nos. (For 10 nos.)
Spec. cl. No: 2807															
Norms no 28.9B	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
	Unskilled	day	0.33	855.00	282.15	seedling	nos	10	3.30	33.00	Add 3 % of Labour cost for Ms rod or hard wood peg and other T&P			8.46	8.46
						compost	cum	0.05	121.53	6.08					
						green mulch	cum	0.04	121.53	4.86					
	Sub total A				282.15	Sub total of B =				43.94	Sub total of C =				8.46
	Sub total of A +B + C =				334.55	Contractor's overhead expenses 15% =				50.18	Total			384.73	
														Unit Rate	38.47

Description of works	Planting containerized tree and shrub seedlings, including pitting, transplanting, composting and mulching, on slopes 30o . - 40o Pit size 30 cm diameter x depth mix Compost with soil and back fill into pit to 1/4 of the pit volume														Unit = nos. (For 10 nos.)
Spec. cl. No: 2807															
Norms no 28.9B	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
	Unskilled	day	0.40	855.00	342.00	seedling	nos	10	3.30	33.00	Add 3 % of Labour cost for Ms rod or hard wood peg and other T&P			10.26	10.26
						compost	cum	0.05	121.53	6.08					
						green mulch	cum	0.04	121.53	4.86					
	Sub total A				342.00	Sub total of B =				43.94	Sub total of C =				10.26
	Sub total of A +B + C =				396.20	Contractor's overhead expenses 15% =				59.43	Total			455.63	
														Unit Rate	45.56

Description of works	Providing and fixing of non reflective warning , mandatory and informatory sign board of 2mm thick MS sheet with back support frame on heavy 50mm tube or Channel section of 75mm *40mm firmly fixed to the ground by means of properly designed foundation with M 10/40 grade cement concrete 300mm *300mm *300mm as per drawings and Technical specifications /DoR Publication.														Unit = nos. (For 4 traffic signs nos.)
Spec. cl. No: 1501															
Norms no 15.1	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
	Unskilled	day	2.00	855.00	1,710.00	excavation for foundation	cum	0.21	106.43	22.35	Tractor with trolly	hr	3.00	1,316.38	3,949.14
	Skilled	day	1.00	1,215.00	1,215.00	Cement concrete M10/40	cum	0.11	9,307.77	1,023.86					
						Paints two coats	sqm	3.51	478.74	1,680.38					
						Mild steel channel iron 75*40*6mm, 3.0 m long @ 6.8 kg/m incl. 5% wastage OR Heavy duty steel tube of intarnal dia 50mm (6.19 kg/m), 3.0 m long including 5% wastage	m	12.6	689.85	8,692.07					
						Angle iron 50*50*6mm for hold fast including 5% wastage	Kg	4.24	159.86	677.81					
						60 cm equilateral triangle	sqm	0.44	2,770.62	1,219.07					
	Sub total A				2,925.00	Sub total of B =				13,315.53	Sub total of C =				3,949.14
	Sub total of A +B + C =				20,189.67	Contractor's overhead expenses 15% =				3,028.45	Total			23,218.12	
														Unit Rate	5,804.53